William R. Gideon
Vice President
Brunswick Nuclear Plant
P.O. Box 10429
Southport, NC 28461
o: 910.457.3698

10 CFR 50.73

June 12, 2017

Serial: BSEP 17-0054

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Subject: Brunswick Steam Electric Plant, Unit Nos. 1 and 2
Renewed Facility Operating License Nos. DPR-71 and DPR-62
Docket Nos. 50-325 and 50-324
Licensee Event Report 1-2017-002

In accordance with the Code of Federal Regulations, Title 10, Part 50.73, Duke Energy Progress, LLC, submits the enclosed Licensee Event Report (LER). This report fulfills the requirement of 10 CFR 50.73(a)(1) for a written report within sixty (60) days of a reportable occurrence.

Please refer any questions regarding this submittal to Mr. Lee Grzeck, Manager – Regulatory Affairs, at (910) 457-2487.

Sincerely,

William R. Gideon

SWR/swr

Enclosure: Licensee Event Report 1-2017-002 cc (with enclosure):

U. S. Nuclear Regulatory Commission, Region II
ATTN: Ms. Catherine Haney, Regional Administrator
245 Peachtree Center Ave, NE, Suite 1200
Atlanta, GA 30303-1257

U. S. Nuclear Regulatory Commission
ATTN: Ms. Michelle P. Catts, NRC Senior Resident Inspector
8470 River Road
Southport, NC 28461-8869

U. S. Nuclear Regulatory Commission
ATTN: Mr. Andrew Hon (Mail Stop OWFN 8G9A) **(Electronic Copy Only)**
11555 Rockville Pike
Rockville, MD 20852-2738
Andrew.Hon@nrc.gov

Chair - North Carolina Utilities Commission **(Electronic Copy Only)**
4325 Mail Service Center
Raleigh, NC 27699-4300
swatson@ncuc.net

NRC FORM 366 (04-2017) **U.S. NUCLEAR REGULATORY COMMISSION**

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. FACILITY NAME	2. DOCKET NUMBER	3. PAGE
Brunswick Steam Electric Plant (BSEP) Unit 1	05000325	1 OF 3

4. TITLE

Foreign Material in Switch Results in Unplanned Automatic Start of Emergency Diesel Generators

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER
									Brunswick Unit 2	05000324
04	17	2017	2017	- 002	- 00	06	12	2017	FACILITY NAME	DOCKET NUMBER 05000

9. OPERATING MODE	11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: *(Check all that apply)*			
1	[ ] 20.2201(b)	[ ] 20.2203(a)(3)(i)	[ ] 50.73(a)(2)(ii)(A)	[ ] 50.73(a)(2)(viii)(A)
	[ ] 20.2201(d)	[ ] 20.2203(a)(3)(ii)	[ ] 50.73(a)(2)(ii)(B)	[ ] 50.73(a)(2)(viii)(B)
	[ ] 20.2203(a)(1)	[ ] 20.2203(a)(4)	[ ] 50.73(a)(2)(iii)	[ ] 50.73(a)(2)(ix)(A)
	[ ] 20.2203(a)(2)(i)	[ ] 50.36(c)(1)(i)(A)	[x] 50.73(a)(2)(iv)(A)	[ ] 50.73(a)(2)(x)
10. POWER LEVEL	[ ] 20.2203(a)(2)(ii)	[ ] 50.36(c)(1)(ii)(A)	[ ] 50.73(a)(2)(v)(A)	[ ] 73.71(a)(4)
	[ ] 20.2203(a)(2)(iii)	[ ] 50.36(c)(2)	[ ] 50.73(a)(2)(v)(B)	[ ] 73.71(a)(5)
	[ ] 20.2203(a)(2)(iv)	[ ] 50.46(a)(3)(ii)	[ ] 50.73(a)(2)(v)(C)	[ ] 73.77(a)(1)
100	[ ] 20.2203(a)(2)(v)	[ ] 50.73(a)(2)(i)(A)	[ ] 50.73(a)(2)(v)(D)	[ ] 73.77(a)(2)(i)
	[ ] 20.2203(a)(2)(vi)	[ ] 50.73(a)(2)(i)(B)	[ ] 50.73(a)(2)(vii)	[ ] 73.77(a)(2)(ii)
		[ ] 50.73(a)(2)(i)(C)	[ ] OTHER Specify in Abstract below or in NRC Form 366A	

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT	TELEPHONE NUMBER *(Include Area Code)*
Lee Grzeck, Manager - Regulatory Affairs	(910) 457-2487

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX
B	TA	33	N007	Y					

14. SUPPLEMENTAL REPORT EXPECTED	15. EXPECTED SUBMISSION DATE	MONTH	DAY	YEAR
[ ] YES *(If yes, complete 15. EXPECTED SUBMISSION DATE)* [x] NO				

ABSTRACT *(Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)*

On April 17, 2017, at 0004 Eastern Daylight Time, Unit 1 was in Mode 1 at approximately 100 percent of rated power, and Unit 2 was in Mode 1 at approximately 22 percent of rated power and was starting up from a refueling outage. Operators manually tripped the Unit 2 main turbine to halt increasing bearing vibration. The power circuit breakers (PCBs) for the Unit 2 main generator did not open as expected on the turbine trip, but subsequently opened when main generator reverse power relays actuated. This resulted in the automatic start of all four emergency diesel generators (EDGs). The EDGs did not tie to emergency busses because offsite power was still available. This event resulted from a component which failed due to foreign material intrusion. A limit switch associated with a main turbine stop valve failed to change states when the turbine was tripped. The limit switch configuration, together with other logic, satisfied the conditions to start the EDGs. The limit switch failure resulted from foreign material in the switch. The failed limit switch was replaced. Planned corrective actions include inspecting similar switches on both units, and sealing the wire entrances of the switch bodies to improve foreign material exclusion.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Brunswick Steam Electric Plant (BSEP) Unit 1	05000325	2017	- 002	- 000

NARRATIVE

Energy Industry Identification System (EIIS) codes are identified in the text as [XX].

Background

Initial Conditions

On April 17, 2017, at 00:04 Eastern Daylight Time (EDT), Unit 1 was in Mode 1 at approximately 100 percent of rated power, and Unit 2 was in Mode 1 at approximately 22 percent of rated power. Unit 2 was restarting following a refueling outage. Personnel were engaged in actions to resolve high bearing vibration on the main turbine [TA].

Reportability Criteria

This event is reportable per 10 CFR 50.73(a)(2)(iv)(A) because the plant experienced a valid start of its four emergency diesel generators (EDGs) [EK].

This event is also reportable per 10 CFR 50.72(b)(3)(iv)(A) via an 8-hour telephone report. The telephone report was completed as Event Number 52683 at 07:40 EDT on April 17, 2017.

Event Description

On April 17, 2017, at 00:04 EDT, the Unit 2 main turbine experienced high bearing vibration. Control Room personnel tripped the main turbine at a prearranged vibration level. The power circuit breakers (PCBs) [FK] for the main generator [EL] did not immediately open on the turbine trip as expected. Subsequently, a main generator lockout occurred on reverse power. The generator lockout sends a start signal to the plant's four EDGs. These started as expected but did not tie to their emergency electrical buses because there was no loss of offsite power. The EDGs were then manually shut down.

Event Cause

This event occurred when a contact in a switch associated with a main turbine stop valve did not close correctly in response to a turbine trip. When the switch was disassembled, a small piece of foreign material was found inside, which was a fragment of a plastic tie-wrap. This interfered with proper contact function. The switch was installed with the wire entrance facing upward which allowed foreign material to fall into the switch body. The apparent cause of the event was that vendor guidance does not require any special foreign material consideration for installation of this switch with its wire entrance facing upward.

Safety Assessment

The site's Electrical Power Distribution System sources consist of the offsite power sources (i.e., preferred and alternate power sources), and the onsite standby power sources, EDGs 1, 2, 3, and 4. The AC electrical power system provides independence and redundancy to ensure an available source of power to

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Brunswick Steam Electric Plant (BSEP) Unit 1	05000325	2017	- 002	- 000

NARRATIVE

the Engineered Safety Feature (ESF) systems. Offsite sources for both units are supplied via the 230 kV switchyard either through a startup transformer or the unit auxiliary transformer. The onsite standby power source for 4.16 kV emergency buses E1, E2, E3, and E4 consists of four EDGs. Each EDG is dedicated to its associated emergency bus. The EDGs are designed to start on indications of a loss of coolant accident or inadequate voltage on the emergency busses. The EDGs also have anticipatory starting signals on certain electrical logic conditions such as main generator reverse power, main generator output breaker failure, startup transformer lockout, and others. Receiving one of these signals results in the engine starting, but the EDGs will not tie to their emergency busses while voltage is adequate from offsite sources.

In this event, the main turbine was manually tripped at a preplanned setpoint to halt an increase in bearing vibration. A reverse power condition resulted in a main generator lockout and a start of all four of the site's EDGs. The EDGs performed as designed given the start signal that was initiated at the time. Since there was no loss of offsite power, all onsite electrical busses remained energized via their normal supplies. Consequently, the EDGs did not tie to their emergency busses and ran unloaded as designed until Operations personnel took action to secure them.

With the exception of the limit switch contact on a main turbine stop valve, all equipment affected by this event performed as expected given the plant condition and logic signals that existed. There was no actual safety consequence that resulted from this event.

Based on this analysis, it is concluded that this event had no adverse impact on nuclear safety.

Corrective Actions

Any changes to the corrective actions and schedules noted below will be made in accordance with the site's corrective action program.

Similarly installed limit switches on the main turbine and bypass valves will be inspected, and seals will be placed at the wire entrance to the switch bodies. This action will be performed during the next scheduled refueling outage on each unit (i.e., Spring 2018 for Unit 1 and Spring 2019 for Unit 2).

Previous Similar Events

No events have been reported in the past three years in which foreign material in a component resulted in an unplanned automatic actuation of a safety-related system or component.

Commitments

This report contains no new regulatory commitments.